STA 445 S26 Assignment 6

2026-03-27

Load your packages here:

Problem 1

A common task is to take a set of data that has multiple categorical variables and create a table of the number of cases for each combination. An introductory statistics textbook contains a data set summarizing student surveys from several sections of an intro class. The two variables of interest are `Gender` and `Year` which are the students gender and year in college. *Note: you will need to refer to Chapter 4 and Chapter 7 for some of the operations needed below - this is a great time to review chapter 4!*

a) Download the data set using the following:

```
Survey <- read.csv('https://www.lock5stat.com/datasets2e/StudentSurvey.csv', na.strings=c('',' ')
```

b) Select the specific columns of interest **Year** and **Gender**

```
Survey <- read.csv('https://www.lock5stat.com/datasets2e/StudentSurvey.csv', na.strings=c('',' ')

(Surv <- Survey %>% select(Year, Gender))
```

```
##          Year Gender
## 1      Senior      M
## 2   Sophomore      F
## 3   FirstYear      M
## 4      Junior      M
## 5   Sophomore      F
## 6   Sophomore      F
## 7   FirstYear      F
## 8   Sophomore      M
## 9      Junior      F
## 10  FirstYear      F
## 11  Sophomore      F
## 12  FirstYear      M
## 13  Sophomore      M
## 14     Junior      F
## 15  FirstYear      M
## 16  FirstYear      F
## 17  FirstYear      F
```

```
## 18  Sophomore      M
## 19  Sophomore      F
## 20  Sophomore      F
## 21  Sophomore      F
## 22  Sophomore      F
## 23  FirstYear      M
## 24  Sophomore      M
## 25  Sophomore      M
## 26  Sophomore      M
## 27  Sophomore      F
## 28     Junior      M
## 29  FirstYear      M
## 30     Senior      M
## 31     Junior      M
## 32  Sophomore      F
## 33  FirstYear      M
## 34     Senior      M
## 35  FirstYear      F
## 36  Sophomore      F
## 37  FirstYear      M
## 38  Sophomore      F
## 39  Sophomore      F
## 40     Junior      F
## 41     Senior      M
## 42  Sophomore      F
## 43  Sophomore      F
## 44  FirstYear      F
## 45  FirstYear      M
## 46     Junior      M
## 47     Senior      M
## 48  Sophomore      M
## 49     Senior      M
## 50  Sophomore      F
## 51  FirstYear      M
## 52     Senior      F
## 53     Junior      M
## 54     Senior      M
## 55     Junior      M
## 56  FirstYear      M
## 57     Junior      F
## 58  Sophomore      F
## 59  Sophomore      F
## 60  FirstYear      F
## 61  Sophomore      M
## 62     Senior      M
## 63  FirstYear      F
## 64  FirstYear      M
## 65  FirstYear      M
## 66  Sophomore      F
## 67  FirstYear      M
```

```
## 68  Sophomore      M
## 69  Sophomore      M
## 70     Senior      M
## 71     Senior      F
## 72  Sophomore      F
## 73  FirstYear      F
## 74  FirstYear      F
## 75  Sophomore      F
## 76  FirstYear      F
## 77  FirstYear      F
## 78  Sophomore      F
## 79  Sophomore      F
## 80  Sophomore      M
## 81  Sophomore      M
## 82  FirstYear      F
## 83  FirstYear      F
## 84     Junior      F
## 85  FirstYear      F
## 86  Sophomore      F
## 87     Senior      M
## 88  FirstYear      F
## 89  FirstYear      M
## 90  FirstYear      M
## 91  Sophomore      M
## 92  FirstYear      F
## 93  Sophomore      F
## 94  Sophomore      M
## 95  FirstYear      M
## 96  Sophomore      M
## 97  Sophomore      F
## 98  Sophomore      M
## 99  Sophomore      M
## 100 FirstYear      M
## 101    Senior      F
## 102 Sophomore      F
## 103 FirstYear      M
## 104 Sophomore      M
## 105 Sophomore      M
## 106      <NA>      F
## 107 Sophomore      M
## 108 Sophomore      M
## 109 Sophomore      F
## 110 Sophomore      M
## 111 Sophomore      M
## 112 Sophomore      M
## 113 Sophomore      F
## 114 FirstYear      M
## 115 FirstYear      F
## 116 Sophomore      F
## 117 Sophomore      F
```

```
## 118 Sophomore      M
## 119 Sophomore      M
## 120 FirstYear      F
## 121 Sophomore      F
## 122 FirstYear      M
## 123 Sophomore      M
## 124 Sophomore      M
## 125 Sophomore      F
## 126 Sophomore      M
## 127 Sophomore      M
## 128 FirstYear      M
## 129 Sophomore      F
## 130 Sophomore      M
## 131    Senior      F
## 132 FirstYear      F
## 133 Sophomore      F
## 134 Sophomore      F
## 135 Sophomore      F
## 136 Sophomore      F
## 137 Sophomore      M
## 138 FirstYear      F
## 139 Sophomore      F
## 140    Junior      F
## 141 FirstYear      M
## 142 FirstYear      F
## 143 FirstYear      M
## 144    Junior      M
## 145 FirstYear      F
## 146 Sophomore      F
## 147 FirstYear      F
## 148 Sophomore      M
## 149 Sophomore      M
## 150 FirstYear      F
## 151 Sophomore      F
## 152 FirstYear      M
## 153 FirstYear      F
## 154 Sophomore      M
## 155    Senior      M
## 156 Sophomore      F
## 157 FirstYear      F
## 158 Sophomore      F
## 159    Senior      F
## 160 Sophomore      F
## 161 FirstYear      M
## 162 FirstYear      F
## 163 Sophomore      M
## 164    Senior      M
## 165 Sophomore      M
## 166 Sophomore      F
## 167    Junior      M
```

```
## 168 FirstYear      M
## 169 Sophomore      M
## 170 Sophomore      M
## 171 Sophomore      F
## 172 Sophomore      F
## 173    Junior      F
## 174 Sophomore      F
## 175    Senior      F
## 176 FirstYear      M
## 177 Sophomore      M
## 178 Sophomore      F
## 179 FirstYear      M
## 180 Sophomore      F
## 181 Sophomore      M
## 182 Sophomore      F
## 183 Sophomore      F
## 184 Sophomore      F
## 185 FirstYear      F
## 186 Sophomore      M
## 187 Sophomore      M
## 188 Sophomore      F
## 189 Sophomore      M
## 190 Sophomore      M
## 191    Junior      M
## 192    Junior      F
## 193 Sophomore      M
## 194 Sophomore      F
## 195 Sophomore      M
## 196 Sophomore      M
## 197 Sophomore      F
## 198 Sophomore      M
## 199 FirstYear      M
## 200 FirstYear      F
## 201 Sophomore      F
## 202 FirstYear      M
## 203    Junior      F
## 204 Sophomore      F
## 205 Sophomore      F
## 206 FirstYear      F
## 207 Sophomore      M
## 208 FirstYear      M
## 209 Sophomore      M
## 210 FirstYear      M
## 211 Sophomore      F
## 212    Senior      M
## 213    Senior      F
## 214 Sophomore      F
## 215 Sophomore      M
## 216 FirstYear      F
## 217 Sophomore      F
```

```
## 218 Sophomore      F
## 219 Sophomore      M
## 220    Junior      M
## 221 Sophomore      M
## 222 Sophomore      M
## 223 Sophomore      M
## 224 Sophomore      M
## 225    Senior      M
## 226 Sophomore      F
## 227 Sophomore      F
## 228 Sophomore      M
## 229 Sophomore      F
## 230 FirstYear      F
## 231 Sophomore      F
## 232 Sophomore      M
## 233 Sophomore      M
## 234 Sophomore      M
## 235    Junior      F
## 236 Sophomore      F
## 237 Sophomore      M
## 238 FirstYear      M
## 239 Sophomore      M
## 240 Sophomore      M
## 241 Sophomore      F
## 242 Sophomore      M
## 243 FirstYear      M
## 244 Sophomore      F
## 245    Senior      M
## 246 Sophomore      F
## 247 FirstYear      M
## 248 Sophomore      M
## 249 Sophomore      M
## 250 Sophomore      M
## 251    Junior      M
## 252 FirstYear      M
## 253 Sophomore      M
## 254    Senior      M
## 255 Sophomore      M
## 256 Sophomore      F
## 257    Junior      F
## 258 Sophomore      M
## 259 Sophomore      M
## 260 Sophomore      M
## 261 Sophomore      M
## 262 FirstYear      M
## 263      <NA>      F
## 264    Junior      M
## 265 FirstYear      F
## 266    Junior      F
## 267 Sophomore      M
```

```
## 268    Junior      M
## 269 Sophomore      M
## 270 Sophomore      F
## 271 FirstYear      M
## 272 FirstYear      F
## 273 Sophomore      F
## 274 Sophomore      F
## 275 FirstYear      F
## 276 Sophomore      F
## 277    Senior      M
## 278    Senior      M
## 279 FirstYear      F
## 280 FirstYear      M
## 281 Sophomore      M
## 282 FirstYear      F
## 283    Senior      M
## 284 Sophomore      F
## 285 Sophomore      F
## 286 Sophomore      F
## 287    Senior      M
## 288 Sophomore      F
## 289 FirstYear      M
## 290 FirstYear      M
## 291 Sophomore      F
## 292 Sophomore      M
## 293    Junior      F
## 294 FirstYear      F
## 295 Sophomore      M
## 296 Sophomore      F
## 297    Senior      M
## 298 Sophomore      F
## 299 FirstYear      M
## 300 Sophomore      M
## 301 FirstYear      F
## 302 Sophomore      M
## 303 Sophomore      M
## 304 Sophomore      M
## 305 Sophomore      F
## 306 Sophomore      M
## 307    Junior      M
## 308    Junior      F
## 309 Sophomore      M
## 310 Sophomore      F
## 311    Senior      M
## 312 Sophomore      M
## 313 FirstYear      F
## 314    Junior      F
## 315 Sophomore      F
## 316 FirstYear      F
## 317    Senior      M
```

```
## 318 Sophomore      F
## 319 Sophomore      M
## 320    Senior      F
## 321    Junior      M
## 322    Senior      M
## 323 FirstYear      M
## 324    Senior      F
## 325    Junior      F
## 326 FirstYear      M
## 327 Sophomore      F
## 328 Sophomore      M
## 329 FirstYear      M
## 330 Sophomore      M
## 331    Junior      M
## 332 FirstYear      M
## 333    Junior      F
## 334 FirstYear      M
## 335    Junior      M
## 336 Sophomore      M
## 337 Sophomore      F
## 338 Sophomore      M
## 339 Sophomore      M
## 340 Sophomore      M
## 341 Sophomore      F
## 342 Sophomore      M
## 343    Senior      M
## 344 Sophomore      M
## 345 FirstYear      M
## 346 Sophomore      F
## 347 Sophomore      M
## 348 FirstYear      M
## 349 FirstYear      F
## 350    Senior      M
## 351 Sophomore      F
## 352 FirstYear      M
## 353 Sophomore      F
## 354 Sophomore      M
## 355    Senior      F
## 356    Junior      F
## 357 Sophomore      M
## 358 Sophomore      F
## 359 Sophomore      M
## 360 Sophomore      F
## 361 FirstYear      M
## 362 Sophomore      F
```

c) Convert the **Year** column to factors and properly order the factors based on common US progression (FirstYear - Sophomore - Junior - Senior)

```r
(Surv <- Surv %>% mutate(Year = factor(Year, levels = c("FirstYear", "Sophomore", "Junior", "Seni
```

```
##          Year Gender
## 1      Senior      M
## 2   Sophomore      F
## 3   FirstYear      M
## 4      Junior      M
## 5   Sophomore      F
## 6   Sophomore      F
## 7   FirstYear      F
## 8   Sophomore      M
## 9      Junior      F
## 10  FirstYear      F
## 11  Sophomore      F
## 12  FirstYear      M
## 13  Sophomore      M
## 14     Junior      F
## 15  FirstYear      M
## 16  FirstYear      F
## 17  FirstYear      F
## 18  Sophomore      M
## 19  Sophomore      F
## 20  Sophomore      F
## 21  Sophomore      F
## 22  Sophomore      F
## 23  FirstYear      M
## 24  Sophomore      M
## 25  Sophomore      M
## 26  Sophomore      M
## 27  Sophomore      F
## 28     Junior      M
## 29  FirstYear      M
## 30     Senior      M
## 31     Junior      M
## 32  Sophomore      F
## 33  FirstYear      M
## 34     Senior      M
## 35  FirstYear      F
## 36  Sophomore      F
## 37  FirstYear      M
## 38  Sophomore      F
## 39  Sophomore      F
## 40     Junior      F
## 41     Senior      M
## 42  Sophomore      F
## 43  Sophomore      F
## 44  FirstYear      F
## 45  FirstYear      M
## 46     Junior      M
```

```
## 47     Senior      M
## 48  Sophomore      M
## 49     Senior      M
## 50  Sophomore      F
## 51  FirstYear      M
## 52     Senior      F
## 53     Junior      M
## 54     Senior      M
## 55     Junior      M
## 56  FirstYear      M
## 57     Junior      F
## 58  Sophomore      F
## 59  Sophomore      F
## 60  FirstYear      F
## 61  Sophomore      M
## 62     Senior      M
## 63  FirstYear      F
## 64  FirstYear      M
## 65  FirstYear      M
## 66  Sophomore      F
## 67  FirstYear      M
## 68  Sophomore      M
## 69  Sophomore      M
## 70     Senior      M
## 71     Senior      F
## 72  Sophomore      F
## 73  FirstYear      F
## 74  FirstYear      F
## 75  Sophomore      F
## 76  FirstYear      F
## 77  FirstYear      F
## 78  Sophomore      F
## 79  Sophomore      F
## 80  Sophomore      M
## 81  Sophomore      M
## 82  FirstYear      F
## 83  FirstYear      F
## 84     Junior      F
## 85  FirstYear      F
## 86  Sophomore      F
## 87     Senior      M
## 88  FirstYear      F
## 89  FirstYear      M
## 90  FirstYear      M
## 91  Sophomore      M
## 92  FirstYear      F
## 93  Sophomore      F
## 94  Sophomore      M
## 95  FirstYear      M
## 96  Sophomore      M
```

```
## 97  Sophomore      F
## 98  Sophomore      M
## 99  Sophomore      M
## 100 FirstYear      M
## 101    Senior      F
## 102 Sophomore      F
## 103 FirstYear      M
## 104 Sophomore      M
## 105 Sophomore      M
## 106      <NA>      F
## 107 Sophomore      M
## 108 Sophomore      M
## 109 Sophomore      F
## 110 Sophomore      M
## 111 Sophomore      M
## 112 Sophomore      M
## 113 Sophomore      F
## 114 FirstYear      M
## 115 FirstYear      F
## 116 Sophomore      F
## 117 Sophomore      F
## 118 Sophomore      M
## 119 Sophomore      M
## 120 FirstYear      F
## 121 Sophomore      F
## 122 FirstYear      M
## 123 Sophomore      M
## 124 Sophomore      M
## 125 Sophomore      F
## 126 Sophomore      M
## 127 Sophomore      M
## 128 FirstYear      M
## 129 Sophomore      F
## 130 Sophomore      M
## 131    Senior      F
## 132 FirstYear      F
## 133 Sophomore      F
## 134 Sophomore      F
## 135 Sophomore      F
## 136 Sophomore      F
## 137 Sophomore      M
## 138 FirstYear      F
## 139 Sophomore      F
## 140    Junior      F
## 141 FirstYear      M
## 142 FirstYear      F
## 143 FirstYear      M
## 144    Junior      M
## 145 FirstYear      F
## 146 Sophomore      F
```

```
## 147 FirstYear      F
## 148 Sophomore      M
## 149 Sophomore      M
## 150 FirstYear      F
## 151 Sophomore      F
## 152 FirstYear      M
## 153 FirstYear      F
## 154 Sophomore      M
## 155    Senior      M
## 156 Sophomore      F
## 157 FirstYear      F
## 158 Sophomore      F
## 159    Senior      F
## 160 Sophomore      F
## 161 FirstYear      M
## 162 FirstYear      F
## 163 Sophomore      M
## 164    Senior      M
## 165 Sophomore      M
## 166 Sophomore      F
## 167    Junior      M
## 168 FirstYear      M
## 169 Sophomore      M
## 170 Sophomore      M
## 171 Sophomore      F
## 172 Sophomore      F
## 173    Junior      F
## 174 Sophomore      F
## 175    Senior      F
## 176 FirstYear      M
## 177 Sophomore      M
## 178 Sophomore      F
## 179 FirstYear      M
## 180 Sophomore      F
## 181 Sophomore      M
## 182 Sophomore      F
## 183 Sophomore      F
## 184 Sophomore      F
## 185 FirstYear      F
## 186 Sophomore      M
## 187 Sophomore      M
## 188 Sophomore      F
## 189 Sophomore      M
## 190 Sophomore      M
## 191    Junior      M
## 192    Junior      F
## 193 Sophomore      M
## 194 Sophomore      F
## 195 Sophomore      M
## 196 Sophomore      M
```

```
## 197 Sophomore      F
## 198 Sophomore      M
## 199 FirstYear      M
## 200 FirstYear      F
## 201 Sophomore      F
## 202 FirstYear      M
## 203    Junior      F
## 204 Sophomore      F
## 205 Sophomore      F
## 206 FirstYear      F
## 207 Sophomore      M
## 208 FirstYear      M
## 209 Sophomore      M
## 210 FirstYear      M
## 211 Sophomore      F
## 212    Senior      M
## 213    Senior      F
## 214 Sophomore      F
## 215 Sophomore      M
## 216 FirstYear      F
## 217 Sophomore      F
## 218 Sophomore      F
## 219 Sophomore      M
## 220    Junior      M
## 221 Sophomore      M
## 222 Sophomore      M
## 223 Sophomore      M
## 224 Sophomore      M
## 225    Senior      M
## 226 Sophomore      F
## 227 Sophomore      F
## 228 Sophomore      M
## 229 Sophomore      F
## 230 FirstYear      F
## 231 Sophomore      F
## 232 Sophomore      M
## 233 Sophomore      M
## 234 Sophomore      M
## 235    Junior      F
## 236 Sophomore      F
## 237 Sophomore      M
## 238 FirstYear      M
## 239 Sophomore      M
## 240 Sophomore      M
## 241 Sophomore      F
## 242 Sophomore      M
## 243 FirstYear      M
## 244 Sophomore      F
## 245    Senior      M
## 246 Sophomore      F
```

```
## 247 FirstYear      M
## 248 Sophomore      M
## 249 Sophomore      M
## 250 Sophomore      M
## 251    Junior      M
## 252 FirstYear      M
## 253 Sophomore      M
## 254    Senior      M
## 255 Sophomore      M
## 256 Sophomore      F
## 257    Junior      F
## 258 Sophomore      M
## 259 Sophomore      M
## 260 Sophomore      M
## 261 Sophomore      M
## 262 FirstYear      M
## 263      <NA>      F
## 264    Junior      M
## 265 FirstYear      F
## 266    Junior      F
## 267 Sophomore      M
## 268    Junior      M
## 269 Sophomore      M
## 270 Sophomore      F
## 271 FirstYear      M
## 272 FirstYear      F
## 273 Sophomore      F
## 274 Sophomore      F
## 275 FirstYear      F
## 276 Sophomore      F
## 277    Senior      M
## 278    Senior      M
## 279 FirstYear      F
## 280 FirstYear      M
## 281 Sophomore      M
## 282 FirstYear      F
## 283    Senior      M
## 284 Sophomore      F
## 285 Sophomore      F
## 286 Sophomore      F
## 287    Senior      M
## 288 Sophomore      F
## 289 FirstYear      M
## 290 FirstYear      M
## 291 Sophomore      F
## 292 Sophomore      M
## 293    Junior      F
## 294 FirstYear      F
## 295 Sophomore      M
## 296 Sophomore      F
```

```
## 297    Senior      M
## 298 Sophomore      F
## 299 FirstYear      M
## 300 Sophomore      M
## 301 FirstYear      F
## 302 Sophomore      M
## 303 Sophomore      M
## 304 Sophomore      M
## 305 Sophomore      F
## 306 Sophomore      M
## 307    Junior      M
## 308    Junior      F
## 309 Sophomore      M
## 310 Sophomore      F
## 311    Senior      M
## 312 Sophomore      M
## 313 FirstYear      F
## 314    Junior      F
## 315 Sophomore      F
## 316 FirstYear      F
## 317    Senior      M
## 318 Sophomore      F
## 319 Sophomore      M
## 320    Senior      F
## 321    Junior      M
## 322    Senior      M
## 323 FirstYear      M
## 324    Senior      F
## 325    Junior      F
## 326 FirstYear      M
## 327 Sophomore      F
## 328 Sophomore      M
## 329 FirstYear      M
## 330 Sophomore      M
## 331    Junior      M
## 332 FirstYear      M
## 333    Junior      F
## 334 FirstYear      M
## 335    Junior      M
## 336 Sophomore      M
## 337 Sophomore      F
## 338 Sophomore      M
## 339 Sophomore      M
## 340 Sophomore      M
## 341 Sophomore      F
## 342 Sophomore      M
## 343    Senior      M
## 344 Sophomore      M
## 345 FirstYear      M
## 346 Sophomore      F
```

```
## 347 Sophomore      M
## 348 FirstYear      M
## 349 FirstYear      F
## 350    Senior      M
## 351 Sophomore      F
## 352 FirstYear      M
## 353 Sophomore      F
## 354 Sophomore      M
## 355    Senior      F
## 356    Junior      F
## 357 Sophomore      M
## 358 Sophomore      F
## 359 Sophomore      M
## 360 Sophomore      F
## 361 FirstYear      M
## 362 Sophomore      F
```

d) Convert the **Gender** column to factors and rename them Male/Female.

```
Surv <- Surv %>% mutate(Gender = factor(as.character(Gender), levels = c("M", "F"), labels = c("M
(unique(Surv$Gender))
```

```
## [1] Male   Female
## Levels: Male Female
```

e) Produce a data set with eight rows and three columns that contains the number of responses for each gender:year combination. *You might want to look at the following functions:* ***dplyr::count*** *and* ***dplyr::drop_na****.*

```
(datset <- Surv %>% drop_na() %>% count(Gender, Year))
```

```
##   Gender      Year  n
## 1   Male FirstYear 51
## 2   Male Sophomore 99
## 3   Male    Junior 17
## 4   Male    Senior 26
## 5 Female FirstYear 43
## 6 Female Sophomore 96
## 7 Female    Junior 18
## 8 Female    Senior 10
```

f) Pivot the table in part (e) to produce a table of the number of responses in the following form:

Gender	First Year	Sophomore	Junior	Senior
Female				
Male				

```r
(datset2 <- datset %>% pivot_wider(names_from = Year, values_from = n, values_fill = 0))
```

```
## # A tibble: 2 x 5
##   Gender FirstYear Sophomore Junior Senior
##   <fct>      <int>     <int>  <int>  <int>
## 1 Male          51        99     17     26
## 2 Female        43        96     18     10
```

Problem 2

For this problem we will consider two simple data sets. Remember to erase eval = FALSE from the code chunk below.

```r
    A <- tribble(
      ~Name, ~Car,
      'Alice', 'Ford F150',
      'Bob',   'Tesla Model III',
      'Charlie', 'VW Bug')

    B <- tribble(
      ~First.Name, ~Pet,
      'Bob',  'Cat',
      'Charlie', 'Dog',
      'Alice', 'Rabbit')
```

a. Squish the data frames together to generate a data set with three rows and three columns. Do two ways: first using `cbind` and then using one of the `dplyr join` commands.

```r
cbind(A, B)
```

```
##      Name             Car First.Name    Pet
## 1   Alice       Ford F150        Bob    Cat
## 2     Bob Tesla Model III    Charlie    Dog
## 3 Charlie          VW Bug      Alice Rabbit
```

```r
inner_join( A, B, by = c("Name" = "First.Name")) %>% select(Name, Car, Pet)
```

```
## # A tibble: 3 x 3
##   Name    Car             Pet
##   <chr>   <chr>           <chr>
## 1 Alice   Ford F150       Rabbit
## 2 Bob     Tesla Model III Cat
## 3 Charlie VW Bug          Dog
```

b. It turns out that Alice also has a pet guinea pig. Add another row to the `B` data set. Do this using either the base function `rbind`, or either of the `dplyr` functions `add_row` or `bind_rows`.

```r
B <- B %>% add_row(First.Name = "Alice", Pet = "Guinea Pig")
```

c. Squish the `A` and `B` data sets together to generate a data set with four rows and three columns. Do this two ways: first using `cbind` and then using one of the `dplyr join` commands. Which was easier to program? Which is more likely to have an error.

cbind was easier to program but also likely to have an error especially with differening number of rows or clumns in the different data sets.

```r
inner_join(A, B, by = c("Name" = "First.Name"))
```

```
## # A tibble: 4 x 3
##   Name    Car             Pet
##   <chr>   <chr>           <chr>
## 1 Alice   Ford F150       Rabbit
## 2 Alice   Ford F150       Guinea Pig
## 3 Bob     Tesla Model III Cat
## 4 Charlie VW Bug          Dog
```

Problem 3

From this book's GitHub there is a .csv file of the daily maximum temperature in Flagstaff at the Pulliam Airport. The link is: https://raw.githubusercontent.com/BuscagliaR/STA_444_v2/master/data-raw/FlagMaxTemp.csv

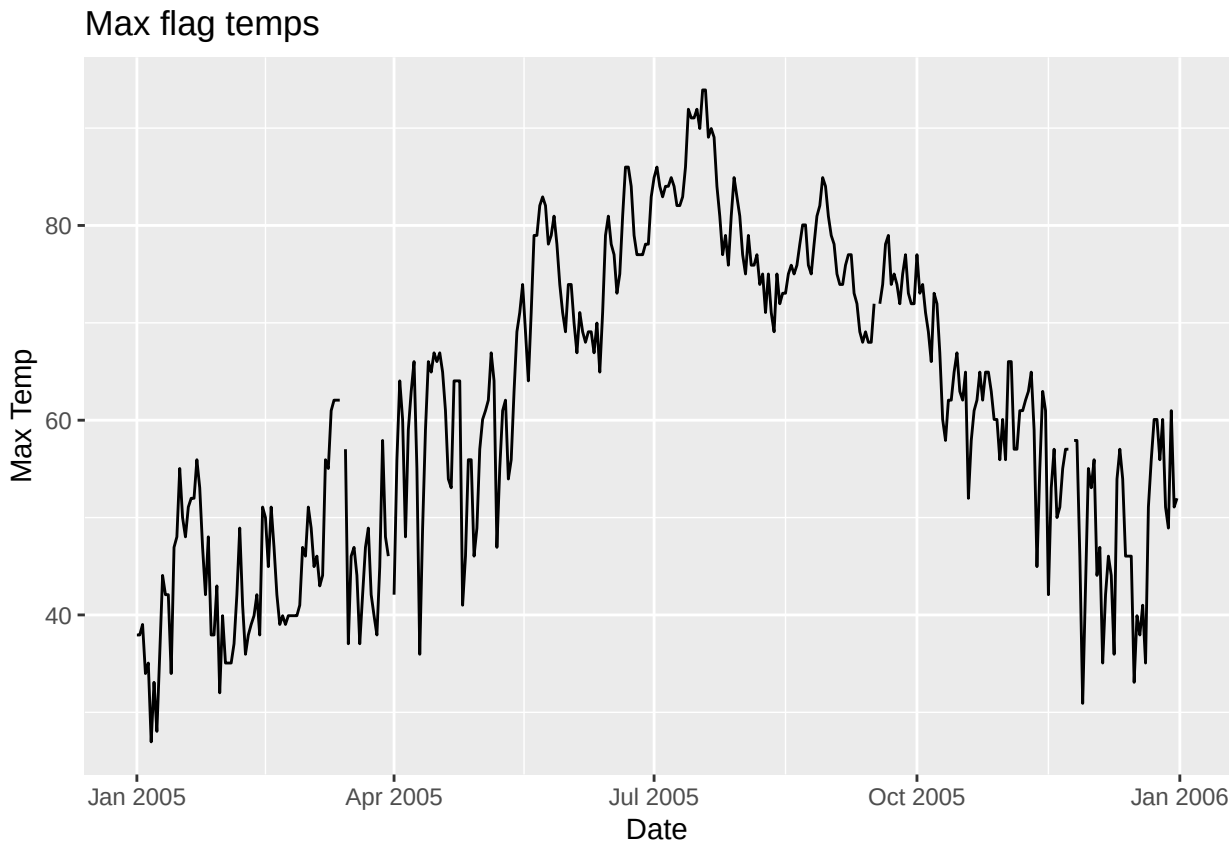
a) Create a line graph that gives the daily maximum temperature for 2005. *Make sure the x-axis is a date and covers the whole year.*

```r
dailytemps <- read_csv("https://raw.githubusercontent.com/BuscagliaR/STA_444_v2/master/data-raw/F
```

```
## New names:
## Rows: 365 Columns: 34
## -- Column specification
## -------------------------------------------------------- Delimiter: "," dbl
## (34): ...1, Year, Month, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, ...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

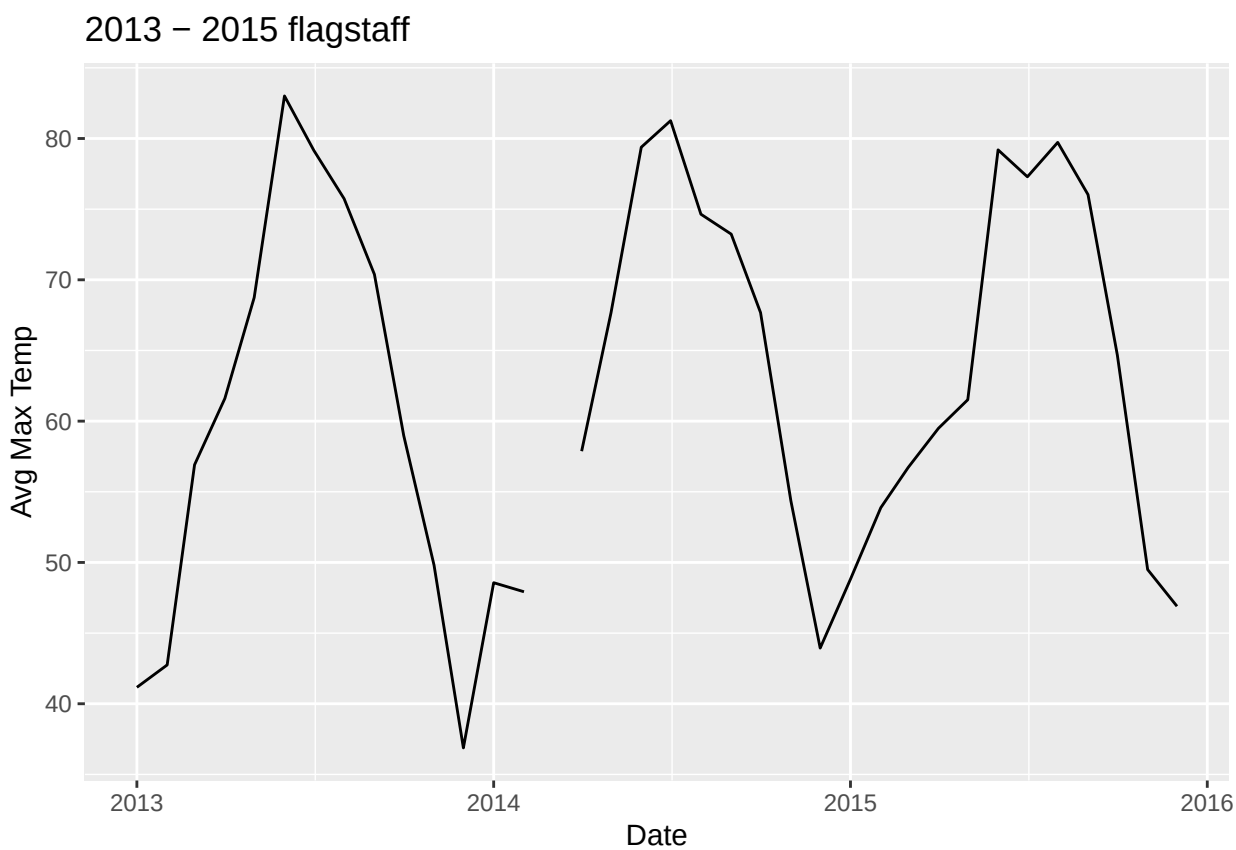
```r
temps_long <- dailytemps %>% pivot_longer(cols = `1`:`31`, names_to = "Day", values_to = "MaxTemp
temps_long <- temps_long %>% mutate(Day = as.numeric(Day), Date = as.Date(paste(Year, Month, Day,
temps_long %>% filter(Year == 2005) %>% ggplot(aes(x = Date, y = MaxTemp)) + geom_line() + labs(t
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



b) Create a line graph that gives the monthly average maximum temperature for 2013 - 2015. *Again the x-axis should be the date and span 3 years.*

```
temps_long %>% filter(Year %in% c(2013, 2014, 2015)) %>% mutate(MonthDate = as.Date(paste(Year, M
  group_by(MonthDate) %>% summarize(AvgTemp = mean(MaxTemp, na.rm = TRUE)) %>% ggplot(aes(x = Mon
  geom_line() + labs(title = "2013 - 2015 flagstaff", x = "Date", y = "Avg Max Temp")
```



Problem 4

The package `nycflights13` contains information about all the flights that arrived in or left from New York City in 2013. This package contains five data tables, but there are three data tables we will work with. The data table `flights` gives information about a particular flight, `airports` gives information about a particular airport, and `airlines` gives information about each airline. Create a table of all the flights on February 14th by Virgin America that has columns for the carrier, destination, departure time, and flight duration. Join this table with the airports information for the destination. Notice that because the column for the destination airport code doesn't match up between `flights` and `airports`, you'll have to use the `by=c("TableA.Col"="TableB.Col")` argument where you insert the correct names for `TableA.Col` and `TableB.Col`.

```
feb14_flights <- flights %>% filter(month == 2, day == 14, carrier == "VX") %>% select(carrier, d
(air_info <- left_join(feb14_flights, airports, by = c("dest" = "faa")))
```

```
## # A tibble: 10 x 11
##    carrier dest  dep_time air_time name        lat   lon   alt    tz dst   tzone
##    <chr>   <chr>    <int>    <dbl> <chr>     <dbl> <dbl> <dbl> <dbl> <chr> <chr>
##  1 VX      LAX        706      347 Los Ange~  33.9 -118.   126    -8 A     Amer~
##  2 VX      SFO        732      344 San Fran~  37.6 -122.    13    -8 A     Amer~
##  3 VX      LAX        909      341 Los Ange~  33.9 -118.   126    -8 A     Amer~
##  4 VX      LAS        934      307 Mc Carra~  36.1 -115.  2141    -8 A     Amer~
##  5 VX      SFO       1029      351 San Fran~  37.6 -122.    13    -8 A     Amer~
##  6 VX      LAX       1317      349 Los Ange~  33.9 -118.   126    -8 A     Amer~
```

```
## 7 VX      LAX       1706      335 Los Ange~  33.9 -118.   126    -8 A     Amer~
## 8 VX      SFO       1746      358 San Fran~  37.6 -122.    13    -8 A     Amer~
## 9 VX      SFO       1852      355 San Fran~  37.6 -122.    13    -8 A     Amer~
## 10 VX     LAX       2017      337 Los Ange~  33.9 -118.   126    -8 A     Amer~
```

Problem 5

Data table joins are extremely common because effective database design almost always involves having multiple tables for different types of objects. To illustrate both table joins and the usefulness of multiple tables we will develop a set of data frames that will represent a credit card company's customer data base. We will have tables for Customers, Retailers, Cards, and Transactions. Below is code that will create and populate these tables.

```r
Customers <- tribble(
  ~PersonID, ~Name, ~Street, ~City, ~State,
  1, 'Derek Sonderegger',  '231 River Run', 'Flagstaff', 'AZ',
  2, 'Aubrey Sonderegger', '231 River Run', 'Flagstaff', 'AZ',
  3, 'Robert Buscaglia', '754 Forest Heights', 'Flagstaff', 'AZ',
  4, 'Roy St Laurent', '845 Elk View', 'Flagstaff', 'AZ')

Retailers <- tribble(
  ~RetailID, ~Name, ~Street, ~City, ~State,
  1, 'Kickstand Kafe', '719 N Humphreys St', 'Flagstaff', 'AZ',
  2, 'MartAnnes', '112 E Route 66', 'Flagstaff', 'AZ',
  3, 'REI', '323 S Windsor Ln', 'Flagstaff', 'AZ' )

Cards <- tribble(
  ~CardID, ~PersonID, ~Issue_DateTime, ~Exp_DateTime,
  '9876768717278723',  1,  '2019-9-20 0:00:00', '2022-9-20 0:00:00',
  '5628927579821287',  2,  '2019-9-20 0:00:00', '2022-9-20 0:00:00',
  '7295825498122734',  3,  '2019-9-28 0:00:00', '2022-9-28 0:00:00',
  '8723768965231926',  4,  '2019-9-30 0:00:00', '2022-9-30 0:00:00' )

Transactions <- tribble(
  ~CardID, ~RetailID, ~DateTime, ~Amount,
  '9876768717278723', 1, '2019-10-1 8:31:23',    5.68,
  '7295825498122734', 2, '2019-10-1 12:45:45',  25.67,
  '9876768717278723', 1, '2019-10-2 8:26:31',    5.68,
  '9876768717278723', 1, '2019-10-2 8:30:09',    9.23,
  '5628927579821287', 3, '2019-10-5 18:58:57',  68.54,
  '7295825498122734', 2, '2019-10-5 12:39:26',  31.84,
  '8723768965231926', 2, '2019-10-10 19:02:20', 42.83)

Cards <- Cards %>%
  mutate( Issue_DateTime = lubridate::ymd_hms(Issue_DateTime),
          Exp_DateTime   = lubridate::ymd_hms(Exp_DateTime) )
Transactions <- Transactions %>%
  mutate( DateTime = lubridate::ymd_hms(DateTime))
```

a) Create a table that gives the credit card statement for Derek. It should give all the transactions, the amounts, and the store name. Write your code as if the only initial information you have is the customer's name. *Hint: Do a bunch of table joins, and then filter for the desired customer name. To be efficient, do the filtering first and then do the table joins.*

```
derekfilter <- Customers %>% filter(Name == "Derek Sonderegger")
(derek_statement <- derekfilter %>% inner_join(Cards, by = "PersonID") %>% inner_join(Transaction
  inner_join(Retailers, by = "RetailID") %>% select(Name.x, DateTime, Amount, Retailer = Name.y))
```

```
## # A tibble: 3 x 4
##   Name.x            DateTime            Amount Retailer
##   <chr>             <dttm>               <dbl> <chr>
## 1 Derek Sonderegger 2019-10-01 08:31:23   5.68 Kickstand Kafe
## 2 Derek Sonderegger 2019-10-02 08:26:31   5.68 Kickstand Kafe
## 3 Derek Sonderegger 2019-10-02 08:30:09   9.23 Kickstand Kafe
```

b) Aubrey has lost her credit card on Oct 15, 2019. Close her credit card at 4:28:21 PM and issue her a new credit card in the `Cards` table. *Hint: Using the Aubrey's name, get necessary CardID and PersonID and save those as* ***`cardID`*** *and* ***`personID`****. Then update the* ***`Cards`*** *table row that corresponds to the* ***`cardID`*** *so that the expiration date is set to the time that the card is closed. Then insert a new row with the* ***`personID`*** *for Aubrey and a new* ***`CardID`*** *number that you make up.*

```
aubrey <- Customers %>% filter(Name == "Aubrey Sonderegger")
personID <- aubrey$PersonID

cardID <- Cards %>% filter(PersonID == personID) %>% pull(CardID)
Cards <- Cards %>% mutate(Exp_DateTime = ymd_hms(Exp_DateTime))
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `Exp_DateTime = ymd_hms(Exp_DateTime)`.
## Caused by warning:
## ! All formats failed to parse. No formats found.
```

```
Cards <- Cards %>% mutate(Exp_DateTime = if_else(CardID == cardID, ymd_hms("2019-10-15 16:28:21")

new_card <- tibble(CardID = "1234567890123456", PersonID = personID, Issue_DateTime = ymd_hms("20
  Exp_DateTime = ymd_hms("2022-10-15 16:28:21"))
Cards <- bind_rows(Cards, new_card)
```

c) Aubrey is using her new card at Kickstand Kafe on Oct 16, 2019 at 2:30:21 PM for coffee with a charge of $4.98. Generate a new transaction for this action. *Hint: create temporary variables* ***`card`****,****`retailid`****,****`datetime`****, and* ***`amount`*** *that contain the information for this transaction and then write your code to use those. This way in the next question you can just use the same code but modify the temporary variables. Alternatively, you could write a function that takes in these four values and manipulates the tables in the GLOBAL environment using the* ***`<<-`*** *command to assign a result to a variable defined in the global environment. The reason this is OK is that in a real situation, these data would be stored in a database and we would expect the function to update that database.*

```
card <- Cards %>% filter(PersonID == aubrey$PersonID & Exp_DateTime > ymd_hms("2019-10-15 16:28:2
retailid <- 1
datetime <- ymd_hms('2019-10-16 14:30:21')
amount <- 4.98
Valid_Cards <- Cards %>% filter(CardID == card, Issue_DateTime <= datetime, datetime <= Exp_DateT

if(nrow(Valid_Cards) == 1){
  new_transaction <- tibble(CardID = card, RetailID = retailid, DateTime = datetime, Amount = amo
  Transactions <- bind_rows(Transactions, new_transaction)
} else {
  print("Card Denied")
}
```

d) On Oct 17, 2019, some nefarious person is trying to use her OLD credit card at REI. Make sure your code in part (c) first checks to see if the credit card is active before creating a new transaction. Using the same code, verify that the nefarious transaction at REI is denied. *Hint: your check ought to look something like this:*

```
card <- '9876768717278723'
retailid <- 2
datetime <- ymd_hms('2019-10-16 14:30:21')
amount <- 4.98

# If the card is currently valid, this should return exactly 1 row.
Valid_Cards <- Cards %>%
  filter(CardID == card, Issue_DateTime <= datetime, datetime <= Exp_DateTime)

# If the transaction is valid, insert the transaction into the table
if( nrow(Valid_Cards) == 1){
  # Some code to insert the transaction
}else{
  print('Card Denied')
}
```

```
card <- cardID
retailid <- 3
datetime <- ymd_hms('2019-10-17 10:00:00')
amount <- 100.00

Valid_Cards <- Cards %>% filter(CardID == card, Issue_DateTime <= datetime, datetime <= Exp_DateT
(if(nrow(Valid_Cards) == 1) {
  new_transaction <- tibble(
    CardID = card,
    RetailID = retailid,
    DateTime = datetime,
    Amount = amount
  )
  Transactions <- bind_rows(Transactions, new_transaction)
} else {
```

```
  print("Card Denied")
})
```

```
## [1] "Card Denied"
```

```
## [1] "Card Denied"
```

e) Generate a table that gives the credit card statement for Aubrey. It should give all the transactions, amounts, and retailer name for both credit cards she had during this period.

```
(aub_statement <- Customers %>% filter(Name == "Aubrey Sonderegger") %>% inner_join(Cards, by = "
  inner_join(Transactions, by = "CardID") %>% inner_join(Retailers, by = "RetailID") %>%
  select(Name.x, DateTime, Amount, Retailer = Name.y) %>% arrange(DateTime))
```

```
## # A tibble: 2 x 4
##   Name.x             DateTime            Amount Retailer
##   <chr>              <dttm>               <dbl> <chr>
## 1 Aubrey Sonderegger 2019-10-05 18:58:57  68.5  REI
## 2 Aubrey Sonderegger 2019-10-16 14:30:21   4.98 Kickstand Kafe
```